A. Inference and Proof Structure

Argument, validity, and countermodel

An **argument** consists of premises and a conclusion. It is **valid** when every valuation or interpretation that makes all premises true also makes the conclusion true. A **countermodel** makes every premise true and the conclusion false; one countermodel proves invalidity.

Propositional rules of inference

$p,\ p \to q \therefore q$	Modus Ponens (MP)
$\neg q,\ p \to q \therefore \neg p$	Modus Tollens (MT)
$p \to q,\ q \to r \therefore p \to r$	Hypothetical Syllogism
$p \lor q,\ \neg p \therefore q$	Disjunctive Syllogism
$p,\ q \therefore p \land q$	Conjunction
$p \land q \therefore p$	Simplification
$p \therefore p \lor q$	Addition
$p \lor q,\ p \to r,\ q \to r \therefore r$	Proof by cases

A propositional derivation

Target: derive $\neg p$ from $p \to q$, $q \to r$, and $\neg r$. Start at $\neg r$ and move backward through the conditional chain:

1.	$p \to q$	premise
2.	$q \to r$	premise
3.	$\neg r$	premise
4.	$\neg q$	MT, 2 and 3
5.	$\neg p$	MT, 1 and 4.

Line 4 removes the possibility of q; line 5 then removes the possibility of p. Name both the rule and the lines used.

Quantifier rules

Universal instantiation (UI): from $\forall x\, P(x)$ infer $P(c)$ for any named object c in the domain.
Universal generalization (UG): after proving $P(c)$ for a completely arbitrary c, infer $\forall x\, P(x)$.
Existential instantiation (EI): from $\exists x\, P(x)$ introduce a fresh name a with $P(a)$. It denotes some witness, not an arbitrary object.
Existential generalization (EG): from $P(c)$ infer $\exists x\, P(x)$.

Universal and existential witnesses differ

Universal route: from $\forall x(P(x) \to Q(x))$ and $\forall x P(x)$, (1) let c be arbitrary; (2) UI gives $P(c) \to Q(c)$ and $P(c)$; (3) MP gives $Q(c)$; (4) because c was arbitrary, UG gives $\forall x Q(x)$.
Existential warning: from $\exists x P(x)$ and $\exists x Q(x)$, EI may introduce different witnesses a and b. You may not conclude $P(a) \land Q(a)$ or $\exists x(P(x) \land Q(x))$ without evidence that the witnesses coincide.

A countermodel for reused witnesses

Build: use domain $\{1, 2\}$; let P hold only of 1 and Q only of 2. **Check premises:** 1 witnesses $\exists x P(x)$, and 2 witnesses $\exists x Q(x)$. **Check conclusion:** neither object satisfies both predicates, so $\exists x(P(x) \land Q(x))$ is false. True premises and a false conclusion prove invalidity.

Invalid forms

$p \to q, q \therefore p$ affirms the consequent. $p \to q, \neg p \therefore \neg q$ denies the antecedent. Neither is valid. A conclusion may be true even when the displayed argument for it is invalid.

A readiness

Can you execute a short derivation with named rules, recognize invalid forms, use UI/EI with correct scopes, and construct a small countermodel when an existential witness has been reused?

B. Direct Proofs from Definitions

Definitions used on this Core

For integers n:

$$\begin{aligned} n \text{ is even} &\iff n = 2k \text{ for some } k \in \mathbb{Z}, \\ n \text{ is odd} &\iff n = 2k+1 \text{ for some } k \in \mathbb{Z}, \\ d \mid n &\iff n = dk \text{ for some } k \in \mathbb{Z}. \end{aligned}$$

A real r is **rational** iff $r = a/b$ for integers a, b with $b \neq 0$; otherwise it is **irrational**. $A \subseteq B$ means $\forall x(x \in A \to x \in B)$.

Direct-proof template

For "if P, then Q":

1. Let the objects be arbitrary subject to the stated domain.
2. Assume P.
3. Unpack P using definitions and introduce correctly scoped witnesses.
4. Calculate or reason to the defining form of Q.
5. State why the new witness belongs to the required domain.

B: an even-square proof

Set up: let arbitrary $n \in \mathbb{Z}$ be even, so $n = 2k$ for some $k \in \mathbb{Z}$. **Calculate and expose a factor of 2:**

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Close: $2k^2 \in \mathbb{Z}$, so the display has form 2(integer). Thus n^2 is even. The new witness is $2k^2$.

B: two odd integers

Set up: let arbitrary odd integers a, b satisfy $a = 2j + 1$ and $b = 2k + 1$ for some $j, k \in \mathbb{Z}$. Use different witnesses. **Combine:**

$$a + b = 2j + 1 + 2k + 1 = 2(j + k + 1).$$

Close: $j + k + 1 \in \mathbb{Z}$, so $a + b$ is twice an integer and is even.

B: sum of rational numbers

Set up: let $r, s \in \mathbb{Q}$. Choose integers a, b, c, d with $b, d \neq 0$, $r = a/b$, and $s = c/d$. **Use a common denominator:**

$$r + s = \frac{ad + bc}{bd}.$$

Close: $ad + bc, bd \in \mathbb{Z}$, and $bd \neq 0$. The displayed fraction satisfies the definition of rational; lowest terms are not required.

B: direct element proof

Assume $A \subseteq B$. **Start on the left:** let arbitrary $x \in A \cap C$. **Unpack:** $x \in A$ and $x \in C$. **Use the hypothesis:** $x \in A$ and $A \subseteq B$ give $x \in B$. **Repack:** $x \in B$ and $x \in C$ give $x \in B \cap C$. Since x was arbitrary, the subset claim follows.

Necessary proof language

Arbitrary means no special property beyond the hypotheses was chosen. A **witness** is a specific object whose existence establishes an existential claim. A **lemma** is a proved supporting result. A **counterexample** satisfies a universal claim's hypotheses but violates its conclusion.

Common errors

Do not write $a = 2k$ and $b = 2k$ merely because both are even; their witnesses may differ. Do not divide by a variable without knowing it is nonzero. Do not give an example when the claim is universal. End by connecting the algebra to the exact definition requested.

B readiness

Can you identify the target definition, introduce independent witnesses, perform the central algebraic or element-chasing step, verify the new witness, and state the conclusion for arbitrary inputs?

C–D. Indirect Methods, Cases, Existence, and Refutation

Contrapositive and contradiction

The contrapositive of $P \to Q$ is $\neg Q \to \neg P$ and is logically equivalent to the original. A **proof by contradiction** assumes the negation of the desired conclusion and derives an impossibility such as $R \wedge \neg R$. State which method you are using.

C: contraposition workflow

Write the contrapositive explicitly, assume its antecedent, and prove its consequent. If the antecedent is a disjunction, use cases. Do not assume the original conclusion.

C: product odd implies both factors odd

Step 1: write the contrapositive: if m is even or n is even, then mn is even. **Step 2:** use cases. If $m = 2k$, then $mn = 2(kn)$; if $n = 2\ell$, then $mn = 2(m\ell)$. **Step 3:** in either case the quantity in parentheses is an integer, so mn is even. The contrapositive, hence the original claim, is true.

C: no largest integer

Assume the opposite: let N be a largest integer. **Construct:** $N + 1$ is an integer and $N + 1 > N$. **Contradiction:** maximality requires $N + 1 \leq N$, while arithmetic gives $N + 1 > N$. Therefore no largest integer exists.

C: rational plus irrational

Let $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$. **Assume the opposite:** $r + x \in \mathbb{Q}$. **Isolate the irrational number:** rationals are closed under subtraction, so $x = (r + x) - r \in \mathbb{Q}$. This contradicts the hypothesis $x \notin \mathbb{Q}$; hence $r + x$ is irrational.

Proof by cases and existence

A **proof by cases** partitions all possibilities and proves the claim in each exhaustive case. A **constructive existence proof** gives an explicit witness and verifies every required property. A **nonconstructive proof** establishes existence without identifying a witness.

D: cases on parity

Exhaustive split: every integer n has form $2k$ or $2k + 1$. **Even case:** $n(n + 1) = 2[k(n + 1)]$. **Odd case:**

$$n(n + 1) = (2k + 1)(2k + 2) = 2[(2k + 1)(k + 1)].$$

Both bracketed quantities are integers. Since all integers occur in one case, $n(n + 1)$ is always even.

D: constructive existence

Witnesses: $24, 25, 26, 27$ are consecutive and positive. **Verification:**

$$24 = 2 \cdot 12, \quad 25 = 5 \cdot 5, \quad 26 = 2 \cdot 13, \quad 27 = 3 \cdot 9.$$

Each factorization is nontrivial, so all four are composite. Listing numbers without checking the required property would not finish the proof.

D: counterexamples

Irrational sum claim: choose $\sqrt{2}$ and $-\sqrt{2}$ (both irrational). Their sum is 0 (rational), so the hypothesis holds and the conclusion fails.
Prime-output claim: choose the allowed input $n = 40$: $40^2 + 40 + 41 = 41^2$, which is composite. One checked counterexample refutes a universal claim.

Common errors

Cases must be exhaustive. A counterexample must obey every domain restriction. In contradiction, do not merely restate the assumed negation; derive a genuine conflict. In contraposition, negate both sides correctly and reverse the arrow.

C–D readiness

Can you state a contrapositive, run a contradiction, split an exhaustive case, verify a constructive witness, and refute a universal claim with a complete counterexample?

E. Strategy Choice and Proof Repair

Choose a promising method

Direct	Definitions turn the hypothesis into the target form.
Contrapositive	The negated conclusion has a useful structural definition.
Contradiction	The conclusion denies existence or involves irrationality.
Cases	The domain naturally partitions (parity, sign, membership).
Constructive	The claim asks only that an example exists.
Counterexample	The universal claim looks too strong.

State the first substantive step, not merely the method name.

E: repair reused witnesses

Gap: "$a = 2k$ and $b = 2k$" forces $a = b$; a shared witness was not justified. **Repair:** choose $j, k \in \mathbb{Z}$ with $a = 2j$ and $b = 2k$. Then $a + b = 2(j + k)$, and $j + k \in \mathbb{Z}$, so the sum is even.

E: repair division by zero

Gap: the original claim fails for $r = 0$, and dividing by r may be illegal. **Correct statement:** if $r \in \mathbb{Q} \setminus \{0\}$ and x is irrational, then rx is irrational. **Proof:** if rx were rational, division by nonzero rational r would give $x = (rx)/r \in \mathbb{Q}$, a contradiction.

E: repair circular reasoning

Gap: assuming n^2 is odd assumes the conclusion. **Repair:** start with the hypothesis $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

The parenthesized quantity is an integer, so the result has form $2(\text{integer}) + 1$ and n^2 is odd.

E: repair a bad starting element

Gap: starting with $x \in A \cap B$ does not cover arbitrary $x \in A$. **Repair:** let arbitrary $x \in A$. The equality $A = A \cap B$ gives $x \in A \cap B$, so $x \in B$. Since x was arbitrary, $A \subseteq B$.

Essential versus minor errors

An **essential error** breaks the proof: assuming the conclusion, reversing an implication, using an unjustified common witness, dividing by a possible zero, omitting a case, or giving an example instead of a universal proof. Minor notation is repairable when scope and reasoning remain unambiguous.

A proof-repair checklist

Ask:

1. Did the proof start with the correct arbitrary object or hypothesis?
2. Are every witness and variable scoped correctly?
3. Is any implication reversed or invalid inference used?
4. Is division known to be legal?
5. Are cases exhaustive and conclusions exact?
6. Does the final line invoke the requested definition?

Name the gap before replacing it.

Presentation matters

A proof is a readable argument, not a stack of equations. State assumptions, connect equations with words, identify the witness produced, and finish with the exact conclusion. A correct idea with a missing essential justification is not a complete proof.

Whole-Core readiness

Can you perform an inference, write a direct proof, use contraposition or contradiction, handle cases/existence/counterexamples, and select or repair a strategy? A live 20-minute form selects one complete problem from each section A–E; four fully correct problems earn Meets and five earn Exceeds.